

Demo: Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones

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Abstract

Cardiovascular diseases remain a leading cause of death worldwide, motivating accessible tools for vascular health monitoring. This demo presents Phonotonos, a smartphone-based system for through-skin ultrasonic sensing of blood-flow velocity waveforms and Doppler indices. Phonotonos repurposes smartphones for vascular sensing despite their low ultrasound transmit power, coarse spatial resolution, and lack of beamforming. The system combines phase-based ultrasonic Doppler sensing with arterial heart sounds and inertial measurements to localize arteries, mitigate motion interference, and recover blood-flow waveforms. Our demonstration showcases an end-to-end prototype in which users position a smartphone over an artery, receive real-time placement guidance, observe the recovered blood-flow waveform, and obtain clinically relevant Doppler indices. Through this interactive demo, we highlight the feasibility of transforming smartphones into portable and accessible tools for everyday vascular screening and monitoring. Additional resources and a demo video are available on our [website](#).

CCS Concepts

• Human-centered computing → Smartphones.

Keywords

acoustic sensing, mobile computing, through-skin sensing, blood flow velocity, mobile health

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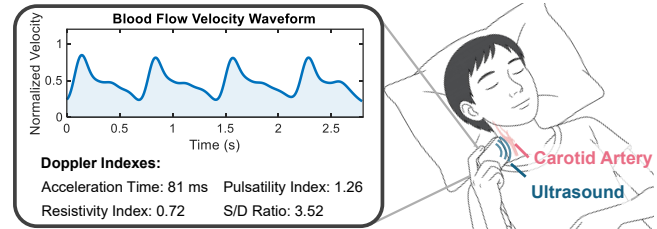


Figure 1: Phonotonos measures blood-flow velocity waveforms and Doppler indices using only a commodity smartphone placed over the carotid artery [3].

1 Introduction

Cardiovascular disease is the leading cause of death worldwide, claiming nearly 18 million lives every year [1, 17]. Clinicians assess vascular health from Doppler blood-flow indices, including Acceleration Time (AT), Systolic-to-Diastolic Ratio (S/D), Resistance Index (RI), and Pulsatility Index (PI), which reveal conditions such as carotid stenosis and peripheral vascular disease [14]. Measuring these indices typically requires costly Doppler ultrasound equipment and trained operators, while handheld devices remain unsuitable for frequent, on-the-go use. This limits routine vascular screening.

We present Phonotonos [3], a system that transforms commodity smartphones into portable tools for through-skin blood-flow sensing. Prior work has shown smartphone-based ultrasonic sensing for physiological monitoring [2, 4, 5, 7, 12] and human-computer interaction [8–11, 15]. Extending this capability to through-skin blood-flow sensing, however, is challenging due to weak Doppler reflections from moving blood cells, low ultrasound transmit power, coarse spatial resolution, and the lack of beamforming [6]. To address these challenges, Phonotonos turns the phone's built-in speaker and microphone into an active ultrasonic sensor. It combines phase-based Doppler sensing, arterial heart sounds, and Inertial Measurement Unit (IMU) measurements in a multi-modal sensing framework to localize arteries, reject motion-corrupted measurements, and recover blood-flow velocity waveforms. From the recovered waveform, Phonotonos derives four clinically relevant Doppler indices.

In this demo, attendees will interact with an end-to-end prototype implementation and experience smartphone-based

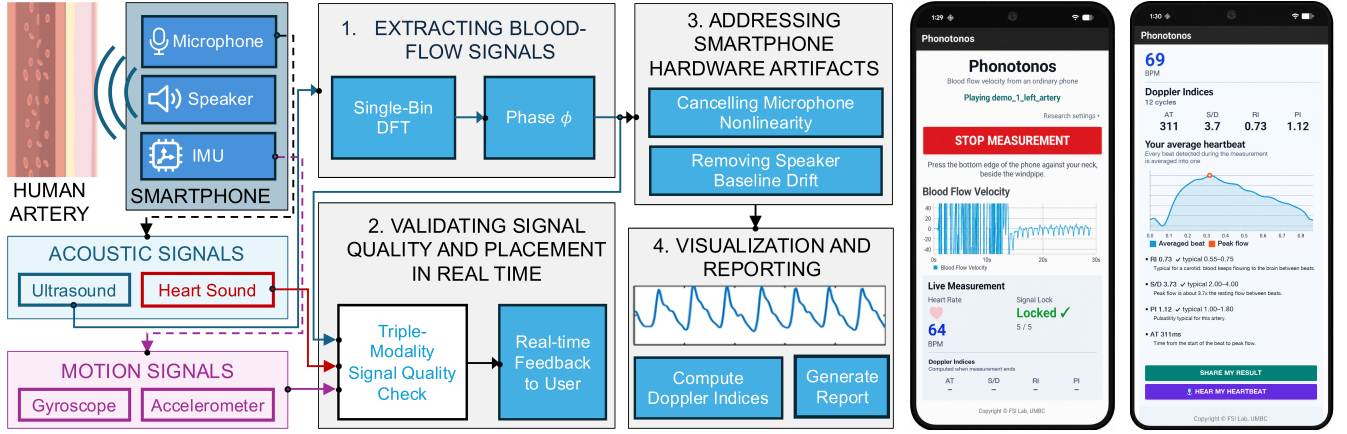


Figure 2: System overview and smartphone interface. Our multi-modal sensing framework integrates ultrasonic Doppler signals, arterial heart sounds, and IMU measurements for real-time artery localization, motion rejection, blood-flow waveform recovery, and on-device Doppler index estimation and visualization.

blood-flow sensing firsthand. As illustrated in Fig. 1, participants place the smartphone over the carotid artery in a comfortable posture, receive real-time guidance for artery localization, and observe the recovered blood-flow velocity waveform and Doppler indices. This demo highlights the potential of smartphones to enable accessible cardiovascular sensing without dedicated medical ultrasound hardware.

2 How Phonotonos Works

Fig. 2 illustrates Phonotonos’ end-to-end pipeline and smartphone interface. During acquisition, the smartphone’s bottom speaker emits a 21 kHz continuous-wave ultrasound signal, while the bottom microphone simultaneously records both the returned ultrasonic signal and arterial heart sound. The phone’s gyroscope and accelerometer provide complementary motion measurements.

Extracting blood-flow signals. We separate the recorded acoustic signal into an audible heart-sound component and an ultrasonic sensing component using band-pass filtering. From the audible band, it extracts the arterial heart-sound waveform and estimates the heart rate for subsequent artery-placement and signal-quality validation. From the ultrasonic band, Phonotonos applies a single-bin Discrete Fourier Transform (DFT) at the transmitted carrier frequency and tracks the corresponding phase variations, which encode the subtle Doppler shifts induced by blood flow. These variations are primarily attributed to moving red blood cells, although arterial wall motion may also contribute. In our full paper, the recovered waveform accurately tracks metered flow in a silicone phantom, while Monte Carlo simulations of red-blood-cell scattering reproduce the observed phase behavior [3].

Validating signal quality and placement in real time. In parallel, Phonotonos uses multimodal sensing to determine whether the phone is correctly positioned over the artery and whether the measurement is reliable. It jointly

uses the phase-based Doppler signal, arterial heart sound, and IMU measurements in a triple-modality signal-quality check. When a reliable arterial signal is detected, the app provides real-time feedback to guide the user toward stable placement, while segments affected by poor positioning or hand motion are rejected.

Addressing smartphone hardware artifacts. Since blood-flow-induced phase variations are subtle, smartphone hardware imperfections can significantly distort the recovered waveform. Phonotonos mitigates the microphone nonlinearity using Least Mean Squares (LMS)-based adaptive cancellation [16] and removes the speaker-induced baseline drift using Baseline Estimation and Denoising with Sparsity (BEADS) correction [13], recovering a normalized blood-flow velocity waveform. From this waveform, AT is measured in milliseconds, while S/D, RI, and PI are computed as ratios independent of the absolute velocity scale.

Cross-platform Visualization and Reporting. We implement Phonotonos on both Android and iOS, with a shared signal-processing pipeline and platform-specific components for audio, IMU access, and user interaction. On both platforms, the app estimates and visualizes the blood-flow velocity waveform and four Doppler indices. As shown in Fig. 2, the app displays the live blood flow velocity waveform and signal-lock status during measurement, then reports the estimated Doppler indices together with a representative cardiac-cycle waveform. Each index is accompanied by its typical range and a brief plain-language explanation for educational purposes rather than clinical diagnosis. All sensing, processing, and reporting run entirely on the phone.

3 Demonstration

Demonstration hardware. We will bring multiple commodity smartphones spanning different vendors, hardware generations, and operating systems, including the Google

Pixel 8 Pro, Pixel 9, Samsung Galaxy S9+, Motorola Edge (2020), iPhone 14, iPhone 14 Pro Max, and iPhone 16. Attendees can perform the same blood-flow measurement on different devices and compare the experience across Android and iOS. The demo runs entirely on-device, requiring no laptop, cloud service, or network connection.

What attendees will do and see. Attendees will place a demo phone over an artery while sitting, standing, or lying down and follow real-time audio and haptic feedback to localize the artery. Once a stable signal is detected, they hold the phone still for approximately 15 s to complete the measurement. The app then visualizes the recovered blood-flow waveform and reports the Doppler indices. The complete interaction takes less than one minute.

Interactive variations. Attendees can explore Phonotonos by measuring different arteries, including the carotid, radial, temporal, and posterior tibial arteries, to observe how blood-flow waveforms vary across body locations. They can repeat measurements across Android and iOS devices from different hardware generations. Attendees can also intentionally move or misplace the phone to observe how the triple-modality signal-quality check detects motion corruption or poor placement and prompts a retry.

4 Validation against Medical Ultrasound

The demo is supported by quantitative validation from our full system paper [3]. In an IRB-approved study with 20 subjects, we performed paired measurements with Phonotonos and a commercial handheld medical Doppler device across four smartphone models and multiple arterial sites. Compared with Doppler ultrasound, Phonotonos achieves 87.5% (AT), 80.4% (S/D), 94.9% (RI), and 81.2% (PI) agreement. The full paper further validates the sensing mechanism through Monte Carlo simulations of red-blood-cell scattering and a silicone phantom driven by a pump and flow meter.

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Generative AI tools were used for language polishing and figure illustration assistance.

References

- [1] British Heart Foundation. 2023. Global Heart & Circulatory Diseases Factsheet. <https://www.bhf.org.uk/what-we-do/our-research/heart-statistics>. Accessed: 2026-07-01.
- [2] Shirui Cao, Dong Li, Sunghoon Ivan Lee, and Jie Xiong. 2023. Powerphone: Unleashing the acoustic sensing capability of smartphones. In *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*. 1–16.
- [3] Shirui Cao, Jie Xiong, Rishav Gupta, Sunghoon Ivan Lee, Jeremy Gummeson, and Dong Li. 2026. [Emerging Ideas] Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones. In *The 24th Annual International Conference on Mobile Systems, Applications and Services (MobiSys '26)* (Cambridge, United Kingdom). ACM, New York, NY, USA. doi:10.1145/3745756.3809246
- [4] Azhar Chara, Tianya Zhao, Xuyu Wang, and Shiwen Mao. 2023. Respiratory biofeedback using acoustic sensing with smartphones. *Smart Health* 28 (2023), 100387.
- [5] Long Fan, Lei Xie, Shiyuan Ma, Yanling Bu, Chuyu Wang, Weibang Pan, Lu Ke, and Sanglu Lu. 2025. SpiroSense: Transforming Smartphones into Pulmonary Metrics Monitors with Ultrasonic Technology. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 9, 3 (2025), 1–26.
- [6] Dong Li. 2026. From Hearing to Feeling: Unlocking Through-Skin Acoustic Sensing on Smartphones. *GetMobile: Mobile Computing and Communications* 29, 4 (2026), 5–10.
- [7] Dong Li, Shirui Cao, Sunghoon Ivan Lee, and Jie Xiong. 2022. Experience: practical problems for acoustic sensing. In *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*. 381–390.
- [8] Dong Li, Jialin Liu, Sunghoon Ivan Lee, and Jie Xiong. 2020. FM-track: pushing the limits of contactless multi-target tracking using acoustic signals. In *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*. 150–163.
- [9] Dong Li, Jialin Liu, Sunghoon Ivan Lee, and Jie Xiong. 2022. Lasense: Pushing the limits of fine-grained activity sensing using acoustic signals. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 1 (2022), 1–27.
- [10] Jialin Liu, Dong Li, Lei Wang, and Jie Xiong. 2021. BlinkListener: "Listen" to Your Eye Blink Using Your Smartphone. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 2 (2021), 1–27.
- [11] Jialin Liu, Dong Li, Lei Wang, Fusang Zhang, and Jie Xiong. 2022. Enabling contact-free acoustic sensing under device motion. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 3 (2022), 1–27.
- [12] Rajalakshmi Nandakumar, Shyamnath Gollakota, and Nathaniel Watson. 2015. Contactless sleep apnea detection on smartphones. In *Proceedings of the 13th annual international conference on mobile systems, applications, and services*. 45–57.
- [13] Xiaoran Ning, Ivan W. Selesnick, and Laurent Duval. 2014. Chromatogram baseline estimation and denoising using sparsity (BEADS). *Chemometrics and Intelligent Laboratory Systems* 139 (2014), 156–167. doi:10.1016/j.chemolab.2014.09.014
- [14] Ammar A. Oqlat, M. Z. Matjafri, Nursakinah Suardi, Mohammad A. Oqlat, Mohannad A. Abdelrahman, and Ahmad A. Oqlat. 2018. A Review of Medical Doppler Ultrasonography of Blood Flow in General and Especially in Common Carotid Artery. *Journal of Medical Ultrasound* 26, 1 (2018), 3–13. doi:10.4103/JMU.JMU_11_17
- [15] Ke Sun, Ting Zhao, Wei Wang, and Lei Xie. 2018. Vskin: Sensing touch gestures on surfaces of mobile devices using acoustic signals. In *Proceedings of the 24th annual international conference on mobile computing and networking*. 591–605.
- [16] Bernard Widrow, John R. Glover, John M. McCool, et al. 1975. Adaptive Noise Cancelling: Principles and Applications. *Proc. IEEE* 63, 12 (1975), 1692–1716. doi:10.1109/PROC.1975.10036
- [17] World Health Organization. 2021. Cardiovascular Diseases (CVDs). [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)). Accessed: 2026-07-01.